

IEC 61558-1:2017 & IEC 61558-2-16:2009+A1:2013 / Annex BB			
Clause	Requirement + Test	Result - Remark	Verdict
BB	Annex BB		P
	Particular requirements for associated transformers for switch mode power supplies with internal frequencies > 500 Hz		P
BB.6	Ratings		P
	Rated output voltage <=50Va.c. or 120V ripple-free d.c. Rated supply frequency <= 500Hz Rated internal operating frequency <=100 MHz		P
BB.8	MARKING AND OTHER INFORMATION		N/A
BB.8.2	Marking for transformers IP00 or for associated transformers: type and trademark, instruction sheets		N/A
BB.8.11	Correct symbols:	Not used for incorporated SMPS	N/A
BB.9	PROTECTION AGAINST ELECTRIC SHOCK	Check together with appliance according to IEC 60335-2-29	N/A
BB.10	CHANGE OF INPUT VOLTAGE SETTING	One input voltage range	N/A
BB.11	OUTPUT VOLTAGE AND OUTPUT CURRENT UNDER LOAD		N/A
BB.12	NO-LOAD OUTPUT VOLTAGE		N/A
BB.13	SHORT-CIRCUIT VOLTAGE		N/A
BB.14	HEATING		P
BB.14.2	Application of 14.1 or 14.3 according to the insulation system		P
BB.14.3	Accelerated ageing test for undeclared class of isolating system		N/A
BB.15	SHORT-CIRCUIT AND OVERLOAD PROTECTION	Test with appliance	N/A
BB.16	MECHANICAL STRENGTH	Test with appliance	N/A
BB.17	PROTECTION AGAINST HARMFUL INGRESS OF WATER AND MOISTURE	Test with appliance	N/A

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BB.18	INSULATION RESISTANCE AND ELECTRIC STRENGTH		P
BB.18.2	Insulation resistance between:		P
	– input circuits and output circuits for double or reinforced insulation $\geq 5 \text{ M}\Omega$		P
	– each input circuit and all other input circuits connected together $\geq 2 \text{ M}\Omega$		P
	– each output circuit and all other output circuits connected together $\geq 2 \text{ M}\Omega$		N/A
BB.18.3	Electric strength test (1 min): no flashover or break-down:		P
	1) double or reinforced insulation between input circuits and output circuits; working voltage (V); test voltage (V)	Working voltage: 289 V; Test voltage: 4098 V	P
	2) basic or supplementary insulation		N/A
	1) Functional insulation for windings intended to be connected in series or parallel (test voltage = working voltage + 500 V) (IEC 61558-2-16:2009)		N/A

BB.19	CONSTRUCTION		P
BB.19.1	Separation of input and output circuits		P
BB.19.1.1	SMPS incorporating auto-transformers (IEC 61558-2-16:2009)	See BB.19.1.3	N/A
BB.19.1.2	SMPS incorporating separating transformers (IEC 61558-2-16:2009)	See BB.19.1.3	N/A
	– Allowed for associated transformers by the equipment standard		N/A
	– Clause 19.8 of part 1 is fulfilled		N/A
BB.19.1.3	SMPS incorporating isolating transformers and safety isolating transformers (IEC 61558-2-16:2009)		P
BB.19.1.3.1	Input and output circuits electrically separated (IEC 61558-2-16:2009)		P
	No possibility of any connection between these circuits		P
BB.19.1.3.2	The insulation between input and output winding(s) consist of double or reinforced insulation (exception see 19.1.3.4) (IEC 61558-2-16:2009)	Triple insulation wire used in secondary winding of transformer	P
	Class I SMPS not intended for connection to the mains by a plug:		—
	Class II SMPS (IEC 61558-2-16:2009)		P

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BB.19.1.3.3	SMPS with intermediate conductive parts not connected to the body (between input/output) (IEC 61558-2-16:2009):		N/A
BB.19.1.3.4	For class I SMPS, with protective screen, not connected to the mains by a plug the following conditions comply (IEC 61558-2-16:2009):		N/A
	Protective screening is not allowed for SMPS with plug connection to the mains (IEC 61558-2-16:2009)		N/A
BB.19.1.3.5	No connection between output circuit and protective earth, except of associated transformers (allowed by equipment standard) or 19.8 is fulfilled (IEC 61558-2-16:2009)		N/A
BB.19.1.3.6	No connection between output circuit and body, except of associated transformers (allowed by equipment standard) (IEC 61558-2-16:2009)		N/A
BB.19.1.3.7	The distance between input and output terminals for the connection of external wiring is 25 mm		N/A
BB.19.1.3.8	Portable SMPS having an rated output ≤ 630 VA (IEC 61558-2-16:2009)		N/A
BB.19.1.3.9	No connection between input and output circuit, except of associated transformers (allowed by equipment standard) (IEC 61558-2-16:2009)		P
BB.19.1.3.10	Protective screening is not allowed for SMPS with plug connection to the mains (IEC 61558-2-16:2009)		N/A
BB.19.12	Windings construction		P
BB.19.12.1	Undue displacement in all types of transformers not allowed:		P
BB.19.12.2	Serrated tape:	Not used	N/A
BB.19.12.3	Insulated windings wires		P
BB.20	COMPONENTS		N/A
BB.21	INTERNAL WIRING		N/A
BB.22	SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS		N/A
BB.23	TERMINALS FOR EXTERNAL CONDUCTORS		N/A
BB.24	PROVISION FOR PROTECTIVE EARTHING		N/A
BB.25	SCREWS AND CONNECTIONS		N/A

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BB.26	CREEPAGE DISTANCES AND CLEARANCES		P
BB.26.1	See 26.101		P
BB.26.2	Creepage distances (cr) and clearances (cr)		P
BB.26.2.1	Windings covered with adhesive tape	Pollution degree 1 not applied	N/A
BB.26.2.2	Uncemented insulating parts pollution degree P2 or P3	Pollution degree P2 applied	P
BB.26.2.3	Cemented insulating parts	No such part	N/A
BB.26.2.4	Enclosed parts, by impregnation or potting	No such construction	N/A
BB.26.3	Distance through insulation	0,2 mm	P
	For double or reinforced insulation, the required values of Tables 13, C1, and D1 – boxes 2b, 2c and 7 are fulfilled		P
BB.26.3.1	Reduced values of the thickness of insulation for supplementary or reinforced insulation are allowed if the following conditions are fulfilled:	Not used	N/A
BB.26.3.2	Insulation in thin sheet form	0,2 mm for reinforced insulation	P
BB.26.3.3	Mandrel test of insulation in thin sheet form (specimen of 70 mm width are necessary):	5000 V for reinforced insulation	P
BB.26.101	Creepage distances, clearances and distances through insulation, specified values according to (IEC 61558-2-16:2009):	6,0 mm for reinforced insulation	P
BB.26.102	Values of IEC 61558-2-16 applicable for frequency up to 3 MHz (IEC 61558-2-16:2009)		P
	For frequency above 3 MHz clause 7 of IEC 60664-4 is applicable (high frequency testing)		N/A
BB.26.103	Clearance (IEC 61558-2-16:2009)		P
	a.) Clearance for frequency ≥ 30 kHz according figure 101 two determinations are necessary:		P
	b.) Clearance for frequency ≤ 30 kHz according figure 101 two determinations are necessary:		N/A
BB.26.104	The working voltages of Table 102, 103 and 104 are peak voltages including μ sec peaks (IEC 61558-2-16:2009)		P
	The working voltage according to Table 13 of part 1 are r.m.s. voltages		P
BB.26.105	Creepage distances		P
	Two determinations of creepage distances are necessary (see Figure 102)		P
BB.26.106	Distance through insulation (IEC 61558-2-16:2009)		P

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	Instead of partial discharge with high frequency voltage the test of the distance and the calculation of the electric field is applicable under the following conditions	Triple insulation wire used	P
BB.26.107	For transformers with FIW wires the following test is required (IEC 61558-2-16:2009/A1:2013)		N/A
BB.27	RESISTANCE TO HEAT, FIRE AND TRACKING	Test with appliance	N/A
BB.E	ANNEX E , GLOW WIRE TEST		N/A
	The test is required according to IEC 60695-2-10 and IEC 60695-2-11 with the following additions:	Test with appliance	N/A
BB.F	ANNEX F, REQUIREMENTS FOR MANUALLY OPERATED SWITCHES WHICH ARE PARTS OF THE TRANSFORMER		N/A
BB.H	ANNEX H, ELECTRONIC CIRCUITS (IEC 61558-1:2017)		N/A
BB.K	ANNEX K, INSULATED WINDING WIRES FOR USE AS MULTIPLE LAYER INSULATION		P
BB.K.1	Wire construction		P
BB.U	ANNEX U – INFORMATIVE – OPTIONAL TW – MARKING FOR TRANSFORMERS		N/A
V	ANNEX V, SYMBOLS TO BE USED FOR THERMAL CUT-OUTS		N/A

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